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Where Minds Pollinate

Computer Science



Class 10

FBISE

Chapter 4

Exercise Solution

Follow Our Socials



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Computer Science 10th

Chapter no 4

Exercise (Q/A)

Q2.1: Describe how data science helps businesses make informed decisions and provide two industry examples.

Data science eliminates guesswork by converting raw data into actionable insights. It uncovers hidden patterns, forecasts market trends, and optimizes operations, allowing companies to make objective, data-driven choices that increase profitability.

Two Industry Examples

1. E-Commerce (e.g., Amazon): Uses predictive algorithms to analyze search history and past purchases. This powers personalized recommendation engines that boost sales and customer retention.

2. Finance (e.g., Banking): Deploys machine learning models to analyze transaction patterns in real-time. This allows institutions to immediately flag and block fraudulent credit card activity.

Q2.2: Identify three ways data science contributes to machine learning and artificial intelligence.

1. Data Cleaning and Preprocessing

Raw data is usually noisy and incomplete. Data science processes clean this data, handle missing values, and remove duplicates to ensure AI models train on high-quality information.

2. Feature Engineering and Selection

AI algorithms require structured inputs to recognize patterns. Data science transforms raw variables into meaningful features, selecting the most relevant data points to improve model accuracy and training speed.

3. Model Evaluation and Validation

Data science provides statistical frameworks to test AI performance. Using metrics like precision, recall, and cross-validation, it ensures models generalize well to new data rather than just memorizing training sets.

Q2.3: Differentiate between supervised learning and unsupervised learning.

Supervised learning	Unsupervised learning
a set of labeled data means that questions with their correct answers are provided to the machine.	unlabeled data is provided to the machine/computer.
After reading/learning a lot of questions/answers the machine is trained.	The algorithm identifies the pattern in the data and divide it into groups based on similarities and differences.
Only new questions without correct answers are provided to that machine and it provides us with the possible correct answers.	When the training is completed, the computer can identify new data and put it in the relevant category.

Q2.4: Describe an everyday example that illustrates reinforcement learning.**Everyday example:**

For example, if you are learning to play chess and whenever you start your game by moving knights (name of a piece in chess game) you win, you will always try to start your game by moving knights. On the other hand, whenever you start your game by moving Queen piece, you lose the game. Therefore, you will always avoid moving Queen piece in the beginning of game in future. In this way, by winning and losing the game by moving certain pieces you have learned winning moves. In reinforcement learning this process is continuous until the model is trained and provides accurate results.

Q2.5: Write the appropriate machine learning model for each of the following scenarios:

Sr no	Scenario	Suitable Machine learning Model
1.	You have a basket of mixed fruits (apple and banana) and you want a robot/machine to sort them.	Supervised Learning
2.	You are given a task to learn how to ride a bicycle to participate in some sports event.	Reinforcement Learning

3.	You have pile of Lego blocks of different colors, and you want your computer to group them by colors regardless of their shapes.	Unsupervised Learning
4.	You have a book with pictures, and you want to teach your sibling to recognize them.	Supervised Learning
5.	You want to train a toy robot to find its way out of maze.	Reinforcement Learning
6.	Your parents want you to clean your messy room if you want to attend the birthday party of your friend.	Reinforcement Learning
7.	You have a set of shapes (square, triangle, circle) and you want to teach a computer to recognize them.	Supervised Learning
8.	You have a book collection without specific categories, and you want your sibling to arrange them according to size, choice or ease of access.	Unsupervised Learning
9.	You are given a task to find the similarity in various flavors of ice cream.	Unsupervised Learning
10.	You have to unlock some rewards in your favorite video game.	Reinforcement Learning

Long Q/A

Q3.1: Analyze the interrelationship between Data Science, Machine Learning, and Artificial Intelligence.

Data Science:

Data Science refers to an interdisciplinary field of mathematics, statistics, and computer science. Main objective of data science techniques is to get meaningful insight from the available data. The available data can be structured as well as unstructured. This objective is achieved through the steps of data collection, data cleaning, data pre-processing and data visualization. Various programming languages like Python, R studio and MATLAB can be used for data visualization.

Artificial Intelligence:

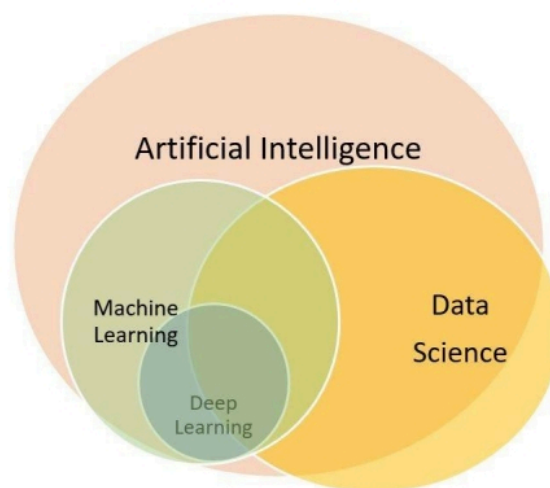
The term Artificial Intelligence is referred to the ability of a machine to exhibit human behavior like problem solving, understanding natural language and interacting with the environment intelligently.

Machine Learning:

Machine learning is used in many fields of life, like healthcare, automation industry, to develop

recommender systems, finance and banking, pattern recognition, NLP, computer vision, research and innovation. In short, the entire scope of Data Science and Artificial Intelligence is based on the algorithms developed in the field of Machine learning.

Data Science, Artificial Intelligence and Machine Learning Interrelationship:



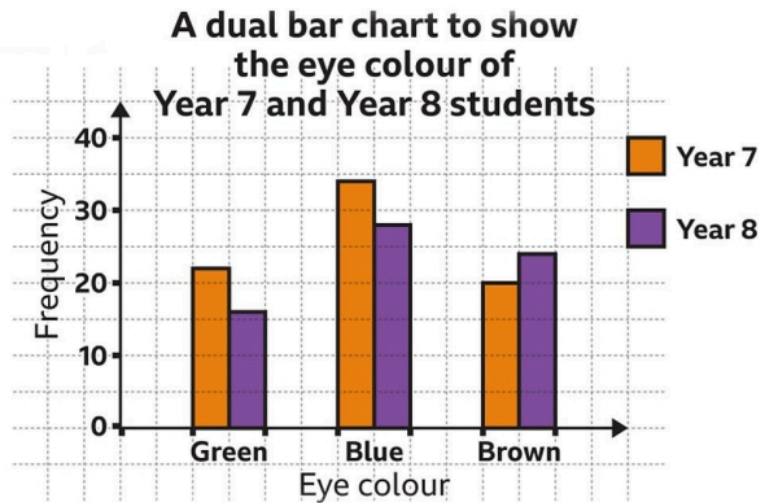
Q3.2: Identify any three types of data visualization, give their applications as well.

Types of Data Visualization

There are different ways to represent data graphically to make it easier to understand. Some common types of data visualization are as follows:

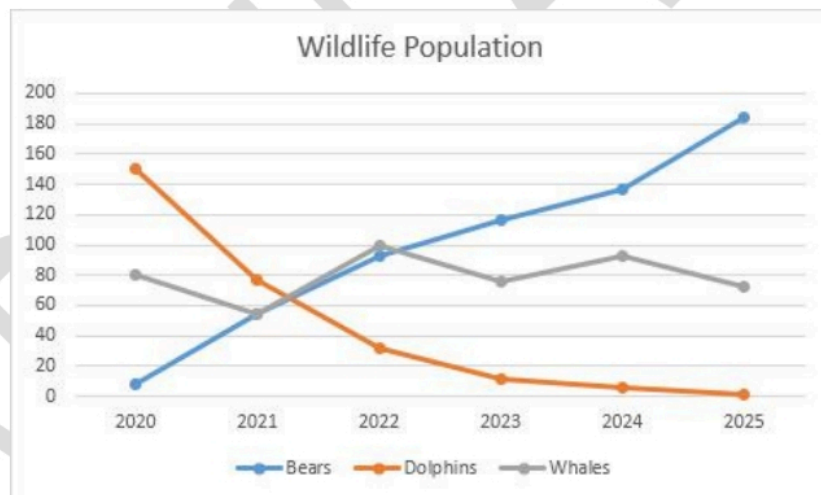
Bar Chart:

This method of data visualization is used to show comparison among discrete values. For example, we can use Bar Chart to visualize marks of students in various subjects or sales data of various products in a company.



Line Chart:

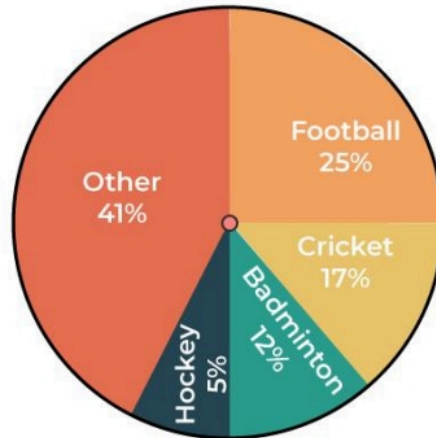
It is used to display trends of data over short or long period of time. For example, gold prices over the last five or ten years can be displayed by using a Line Chart. Similarly, stock prices, or change in temperature over a year can be presented by using a Line Chart.



Pie Chart:

It is used to represent proportions of various categories of data in a whole. For example, market shares of various companies.

Number of Students



Q3.3: Discuss the way data visualization can be used to communicate data uncertainty, provide two specific examples.

Data uncertainty refers to the inherent degree of error, noise, variance, or incompleteness present within a dataset. It arises from limitations in scientific measurement tools, sampling errors, or predictive modeling variances. Communicating this uncertainty ensures that charts do not project a false sense of absolute accuracy, allowing stakeholders to make realistic, risk-aware decisions.

Two specific examples include:

1. Error Bars on a Bar Chart

A thin T-shaped vertical or horizontal line is layered over a primary chart marker to encode a secondary data dimension (such as standard deviation or a 95% confidence interval). A shorter bar indicates tight data collection with high confidence, while a longer line alerts the viewer to significant dispersion and lower precision.

Example: Displaying public opinion margins during **election polls** where candidates are listed alongside their percentage brackets with a $\pm 3\%$ margin of error.

2. Confidence Bands around a Line Graph

Rather than individual markers, a continuous semi-transparent shaded ribbon or "envelope" envelops the primary trendline. The width of the shaded area displays the uncertainty framework at every progressive step across a time series.

Example: Illustrating future **weather and climate forecasts** where a solid path displays the

estimated average temperature while a widening colored band displays the realistic upper and lower limits of potential climate shifts.

Q3.4: Describe the key considerations in selecting appropriate visualizations for different types of data and analyses?

There are different ways to represent data graphically to make it easier to understand. Some common types of data visualization are as follows:

Quantitative Visualization

Quantitative visualization is used to represent numerical data. It focuses on quantities or numbers to show measurable data. It is used to display data that can be measured or counted. For example, a bar chart showing sales figures of a company over several months effectively communicates quantitative information.

Categorical Visualization

Categorical visualization is used to represent data that falls into distinct categories. It helps to show proportions or parts of a whole. This type is ideal for displaying nominal or ordinal data. For example, a pie chart showing the market share of different companies in a specific industry.

Temporal Visualization

Temporal visualization is used to display data that changes over time. It is used for time-series data. Line graphs are usually used to represent temporal data. For example, a line graph.

Spatial Visualization

Spatial visualization is used to represent data related to physical locations or spaces, visualizing geographic or spatial data. For example, a heatmap showing population density across different regions.

Multivariate Visualization

Multivariate visualization is used to represent data involving more than two variables or dimensions. Scatter plots and heatmaps are effective tools to represent such data. For example, a scatter plot matrix showing relationships between income, age, and spending habits.

Interactive Visualization

Interactive visualization allows users to interact with the data, through digital platforms. The digital platforms can be dashboards and filters. A dashboard helps the user to filter and manipulate data

visualizations and explore different trends and insights.

Statistical Visualization

Statistical visualization is used to present data to show statistical properties, such as distribution or correlation. Histograms, box plots, and scatter plots are popular choices.

Information Visualization

Information visualization is used to present complex data sets in an easier way, often for abstract or conceptual data. Network diagrams, tree maps, and word clouds are effective tools. For example, a network diagram showing relationships between entities, such as a social network.

Q3.5: Explain the uses of data visualization in detail.

Uses of Data Visualization

Like Artificial Intelligence, data science and machine learning, data visualization is useful in almost all the fields of life. Some of them are as follows:

Business Intelligence:

Data visualization helps to make data driven, well informed decisions. It is used to find market trends and helps to track and improve performance.

Healthcare:

It helps to visualize the impact of various diseases affecting the patient. It is helpful to track disease and visualize the spread of disease.

Education:

Data visualization is very helpful to teach data literacy skills, concept building, creative thinking, and critical thinking.

Science and Research:

It is useful to visualize complex findings, very huge and complex data such as complex scientific data received from satellite in the form of photographs.

Sports and gaming:

It is useful to visualize performance of players whether they are playing football in a ground or chess players playing in online tournaments. It is also helpful in sports broadcasting, and other predictions.

Finance:

It is helpful to analyze market trends, to track portfolio performance and to identify investment opportunities.

Entertainment:

It helps the entertainment industry, to visualize performance data to predict future trends. It helps in content optimization by visualizing audiences insights and trends.

Q3.6: What are the potential consequences of poor data quality on model performance?

1. Conceptual Principle: "Garbage In, Garbage Out"

The foundational law of machine learning dictates that an algorithm's output is only as reliable as its input data. Poor data quality disrupts the mathematical learning process. When a model trains on corrupted, noisy, or incomplete datasets, the algorithm creates flawed internal rules. This leads to operational failures, high error rates, and unreliable predictions in production.

2. Statistical Misalignment: Bias-Variance Dilemma

- **Underfitting due to High Bias:** If data contains excessive noise or random missing records, the algorithm fails to distinguish the true signal from the errors. It oversimplifies its mathematical assumptions, leading to high bias. The model underfits, failing to achieve acceptable accuracy even on the training dataset.
- **Overfitting due to High Variance:** When data contains uncleaned anomalies, extreme outliers, or duplicate inputs, highly flexible algorithms (like Deep Neural Networks) memorize these specific flaws. The model mistakes this noise for valid patterns. It achieves near-perfect training accuracy but fails completely when exposed to new, unseen data (high variance).

3. Operational Consequences on Performance Metrics

- Poor data quality directly degrades standard evaluation metrics, rendering the system ineffective:
- **Drop in Accuracy and Precision:** Mislabeled data points blur the classification boundaries. For instance, if fraudulent transactions are mislabeled as legitimate, the model's precision drops, creating dangerous false negatives.
 - **Skewed Recall/Sensitivity:** Imbalanced data generation (e.g., a dataset where 99% of entries belong to a single class) tricks the model into always predicting the majority class. The model appears accurate on paper, but its recall for the minority class drops to zero.

4. Algorithmic Bias and Structural Skew

When training data suffers from sampling or historical bias, the machine learning model permanently

embeds and amplifies these human prejudices. For example, if an automated hiring AI is trained on historical data where male applicants were systematically favored, the algorithm learns to penalize female candidates based on their features, resulting in unethical and discriminatory automation.

5. Technical Model Decay and Data Drift

Data quality issues caused by inconsistent formatting—such as mismatched date syntax, changing units (meters vs. feet), or broken API data pipelines—lead to structural data drift. Because the deployment model relies on strict matrix mathematics, unexpected format changes disrupt its numerical calculation pipelines, causing immediate model decay and system crashes.